COMP34812 Natural Language Understanding Project

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INTRODUCTION

- task :
  - Natural Language Interence
- Problme :
  - Given two sentences - a premise and a hypothesis, label them as either "entailment" (1) or "non-entailment" (0) depending on the logical relation implied by them.
- Motivation :
  - Develop models that can verify logical factual consistency, such as for natural text databases or Q&A pipelines.
- Approaches used :
  - 1 .Support Vector Machine (SVM) for Traditional ML (Category A)[2]
  - 2. RoBERTa Large which is a high-performance language model based on BERT, trained on more data with 355M parameters[3].
- Both approaches showed significant results compared to the baseline models evaluated o the development set

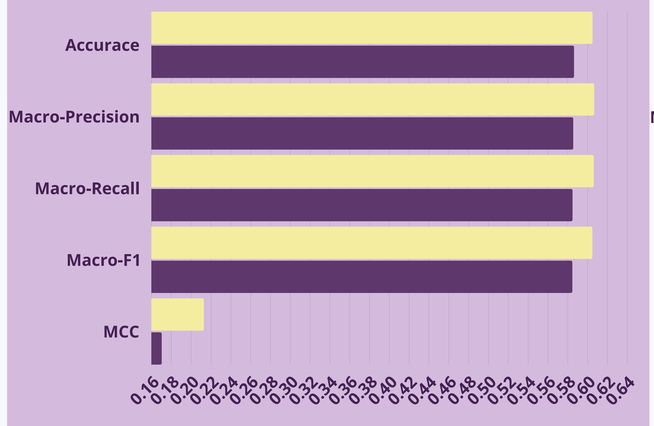
Premise	Hypotesis	Label
A man is playing a guitar on stage in front of an audience.	A person is performing music for others.	1
A woman is sitting at a desk writing in a notebook.	She is working as a teacher.	0

METHODOLOGY

- Category A: SVM with TF-IDF + WordNet.
  - We built a traditional ML pipeline using:
    - TF-IDF vectorization of the combined premise and hypothesis ([SEP] token used).
    - Semantic features from WordNet: synonym overlap and antonym count.
    - Features were concatenated and passed to a linear SVM classifier with class balancing.
- Category C: RoBERTa Fine-Tuning
  - We fine-tuned a RoBERTa-large model for sequence classification:
    - Premise-hypothesis pairs tokenized using the RoBERTa tokenizer.
    - Model includes 24 transformer layers and a classification head.
    - Trained with cross-entropy loss, evaluated using F1, Accuracy, and MCC.

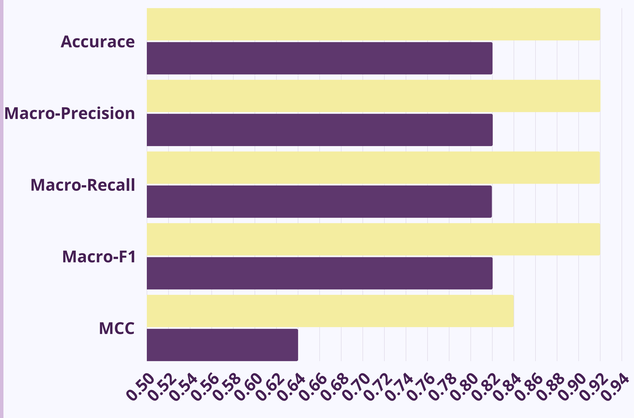
RESULTS

Category (A)



	Accuracy	Macro-P	Macro-R	Macro-F1	MCC
Our SVM	0.6050	0.6066	0.6062	0.6048	0.2128
BL SVM	0.5863	0.5855	0.5848	0.5857	0.1703

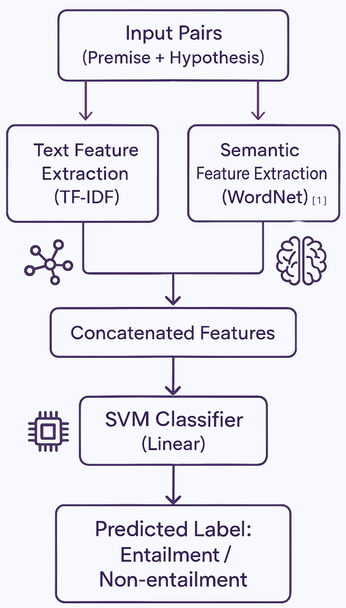
Category (C)



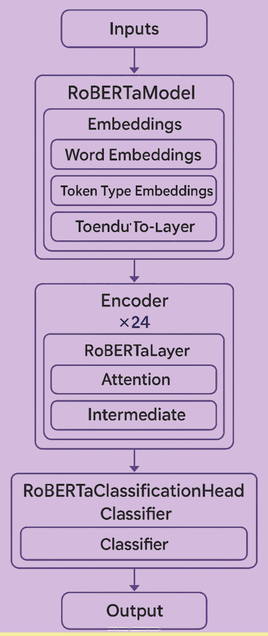
	Accuracy	Macro-P	Macro-R	Macro-F1	MCC
Our Bert	0.92	0.9202	0.9196	0.9199	0.8398
BL Bert	0.8202	0.8204	0.8196	0.8198	0.64

Architecture

Category (A)



Category (C)



CONCLUSION

- The experiments highlight how effectively these approaches can address the Natural Language Inference (NLI) task.
- On the development set, our SVM and RoBERTa models achieved improvements of approximately 2% and 10%, respectively, in Accuracy, Recall, and F1-Scor.
- Future Work
  - Domain-Specific Fine-Tuning: Fine-tune RoBERTa on domain-specific NLI datasets (e.g., medical or legal) to boost performance in specialized areas[4].
  - Model Compression: Apply techniques like distillation or quantization to create a lighter RoBERTa variant for real-time applications[5].

REFERENCES

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3-Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L. and Stoyanov, V., 2019. RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint arXiv:1907.11692.

4-Howard, J. and Ruder, S., 2018. Universal language model fine-tuning for text classification (ULMFiT). In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp.328–339.

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